

Notes: Allen, Qian, Shan, and Zhu (JF, 2024)

Dissecting the Long-Term Performance of the Chinese Stock Market

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1 Big picture

- **Question.** Why has China’s domestic A-share stock market performed so poorly over the long run, despite China’s extraordinary macroeconomic growth, and why does it underperform not only global peers but also externally listed Chinese firms and matched unlisted Chinese firms?
- **Setting.** The paper studies the Chinese stock market over the period **1991–2018**, with most firm-level tests focused on **2000–2018**. It compares four groups:
 1. **A-share firms** listed in Shanghai and Shenzhen,
 2. **externally listed Chinese firms** listed mainly in Hong Kong and the United States,
 3. **listed firms from other countries**, and
 4. **matched unlisted Chinese firms** from manufacturing industries.
- **Core challenge.** Low stock returns can arise for very different reasons:
 - poor fundamentals and weak cash-flow growth,
 - high valuations caused by investor optimism,
 - low discount rates because of financial repression,
 - selection distortions created by IPO and delisting rules,
 - weak corporate governance after listing.

The paper therefore tries to explain **both stock returns and accounting performance** in one framework.

- **Main theoretical contribution.** The authors build a simple model in which investors value A-share firms under the possibility of future market-wide institutional and governance reforms. The model has three versions:
 1. a rational model with common beliefs about reform probabilities;
 2. a behavioral model with optimistic and realistic investors under short-sale constraints;
 3. a segmented-markets model with financial repression.

These versions generate three hypotheses: **institutional deficiencies**, **behavioral biases**, and **financial repression**.

- **Main empirical contribution.** The paper combines:
 - a new comprehensive data set of **externally listed Chinese firms**,
 - cross-country comparisons,
 - within-China comparisons to **matched unlisted firms**,
 - IPO event-window evidence,
 - governance and sentiment measures,
 - delisting and “ST” evidence,
 - and robustness checks on discount rates, risk, and valuation.

- **Why it matters.** China's domestic stock market is the **second largest in the world by market capitalization**. If the market allocates capital poorly or prices firms irrationally, the consequences matter for:
 - firm financing and investment,
 - household saving,
 - innovation,
 - and aggregate resource allocation.
- **Main empirical question.** Are the poor long-run returns of A-share firms mainly explained by:
 1. weak listing/delisting institutions and weak governance,
 2. investor sentiment and speculative pricing,
 3. or low discount rates due to financial repression?
- **Main empirical results.**
 - Over **2000–2018**, if an investor put **RMB 1** in a value-weighted A-share portfolio, the real value at the end of 2018 would be only about **RMB 1.01**.
 - By contrast, externally listed Chinese firms deliver **about 344%** real buy-and-hold returns.
 - In cross-country WLS regressions, A-share firms underperform listed firms from other countries by about **15 percentage points per year**.
 - Externally listed Chinese firms perform roughly on par with, or better than, global peers.
 - A-share firms invest **more** than comparable firms but generate **lower net cash flows**.
 - The underperformance is concentrated in **large A-share firms**; the smallest 30% of A-share firms do not underperform in either stock or accounting returns.
 - IPO rules and weak delisting rules create **adverse selection** into the listed market.
 - Investor **sentiment** is a major predictor of low subsequent returns.
 - **Corporate governance** helps explain both stock returns and accounting performance.
 - **Financial repression** can explain part of the return gap relative to other emerging markets, but it cannot explain the poor accounting performance of A-share firms or the superior performance of externally listed Chinese firms.
- **Mechanism.** The paper argues that the A-share market suffers from a combination of:
 - distorted entry and exit,
 - weak value-maximizing governance,
 - overoptimistic pricing by domestic investors,
 - and only secondarily from low discount rates.
- **Economic takeaway.** A large stock market can still perform poorly if listing rules, exit rules, governance, and investor composition distort both **which firms get listed** and **how those firms are priced and governed after listing**.

2 Data and measurement (key objects)

2.1 Cross-country and Chinese data sources

- **Cross-country data.**
 - Country-level macro variables come from the **World Bank**.
 - Institutional variables come from the **law-and-finance literature**, including investor protection and legal environment measures.
 - Exchange-level variables come from the **World Federation of Exchanges**.
 - Firm-level stock returns and accounting data come from **Worldscope/Datastream**; for the United States and a few other markets the paper also uses **Compustat** and **CRSP**.
 - Governance variables for the cross-country sample come from **BoardEx** and **Risk Metrics**.
 - Investor sentiment variables follow **Baker, Wurgler, and Yuan (2012)**.
- **Cross-country sample size.** The data set contains more than **113,000** unique firms, **157** exchanges, and **106** countries over **1991–2018**.
- **Chinese-specific data.**
 - A-share stock prices and returns come from the **Chinese Capital Market Research Group** and **WIND**.
 - Ownership structure, governance, and related-party transaction information come from **WIND** and **CSMAR**.
 - The **Chinese Industrial Enterprises Database (CIED)** provides unlisted manufacturing firms for **1998–2013**.
- **Main analysis window.** Although the broader sample starts in 1991, most firm-level analysis focuses on **2000–2018**, when the market is more mature and the pace of new listings is more stable.

2.2 Firm groups and sample construction

- **A-share firms.** These are firms listed on the **Shanghai** and **Shenzhen** stock exchanges.
- **Externally listed Chinese firms.** These are firms headquartered in mainland China but listed outside the mainland, mainly in:
 - **Hong Kong**,
 - the **United States**,
 - **Singapore**,
 - and a small set of other markets.
- **Counts.**
 - There are **3,695** unique A-share firms.
 - There are **1,770** unique externally listed Chinese firms.

- Among the external firms, **1,138** are listed in Hong Kong and **515** in the United States.
- **SOEs.** State ownership is defined using ultimate control. In 2000, about **70%** of listed A-share firms are SOEs; by 2018 this falls to about **30%**.
- **Firm size.** Externally listed Chinese firms are much larger on average:
 - average assets of A-share firms over the sample are about **\$1.59 billion**,
 - average assets of externally listed Chinese firms are about **\$8.35 billion**.
- **Matched unlisted firms.** For manufacturing A-share firms, the paper matches listed firms to unlisted firms in the same industry with similar asset size. This yields **1,407** matched A-share firms.
- **Matched externally listed Chinese firms.** The paper also constructs industry-and-size-matched samples comparing A-share firms with externally listed Chinese firms.
- **Cross-listed firms.** A subset of **99** firms are cross-listed in both the A-share market and Hong Kong; these are used in discount-rate and robustness exercises.

2.3 Key variables and measured objects

- **Stock performance.**
 - annual real stock returns,
 - value-weighted buy-and-hold returns (BHRs),
 - correlations between stock returns and future GDP growth.
- **Accounting performance.**
 - **ROA** = EBIT / Total Assets,
 - **ROE** = Net Income / Equity,
 - **ROS** = (Sales – Cost of Goods Sold) / Sales,
 - **Operating Cash Flows (OCF)** = (EBITDA – ΔWC – Taxes)/Assets,
 - **Net Cash Flows** = (OCF – Capital Expenditures)/Assets,
 - **Investment** = Capital Expenditures / Total Assets,
 - **ROIC** = net operating profits divided by invested capital.
- **IPO-related variables.**
 - changes in **ROA** and **ROS** around IPO,
 - **Total Accruals** = Net Income – OCF,
 - IPO first-day returns,
 - years since IPO.
- **Delisting and distress.**
 - **ST** (special treatment) status in China after sustained losses,

- U.S. delisting due to poor performance,
- permanent ST status as a proxy for firms that would likely be delisted in stricter markets.
- **Governance measures.**
 - **G-Index-I** for the cross-country sample:
 - * CEO duality,
 - * board size,
 - * ownership concentration,
 - * executive holdings.
 - **G-Index-A** for A-share firms:
 - * the four items above,
 - * plus state ownership,
 - * plus related-party transaction (RPT) outflows.
- **Behavioral measures.**
 - **Sentiment1**: based on volatility premium and turnover,
 - **Sentiment2**: based on volatility premium, turnover, IPO counts, and IPO returns,
 - **AH premium**: price premium of A-shares over H-shares for dual-listed firms,
 - **stock-level turnover**: a proxy for speculative trading.

2.4 Key descriptive facts

- **A-share stock performance is extremely weak.** A value-weighted A-share portfolio essentially earns **zero real return** from 2000 to 2018.
- **Externally listed Chinese firms perform very well.** Their cumulative real buy-and-hold return is around **344%** by the end of 2018.
- **The poor A-share performance is driven by large firms.** The smallest 30% of A-share firms do not exhibit the same underperformance.
- **The domestic stock market is not a good barometer of the Chinese economy.** The correlation between five-year A-share stock returns and future five-year GDP growth is about **-1.3%** and statistically insignificant; the corresponding correlation for externally listed Chinese firms is about **41%**.
- **A-share firms invest more but generate less cash.** Relative to other groups, they look like **high-investment, low-cash-flow** firms.
- **Low returns are not compensated by better income alternatives for households.** For most of the sample, cumulative A-share returns are below returns on rolling bank deposits and government bonds.

3 Models and empirical specifications (all equations + interpretation)

3.1 Timeline and state variables

- The model is an **infinite-horizon, discrete-time** model with risk-neutral investors.
- Initially, the institutional and governance environment is in a **low state**, with cash-flow growth rate g_L .
- With probability π , comprehensive reforms are announced and implemented, moving the firm into a **high state** with growth rate g_H .
- The opportunity cost of capital is r , with

$$r > g_H > g_L.$$

Interpretation. The model treats A-share firms as potentially low-growth firms today whose value depends on whether investors expect major improvements in the institutional environment.

3.2 Baseline model with common investor beliefs

After reforms are announced and implemented at date t , the firm's value is

$$V_{H,t} = \frac{EC_{L,t}(1 + g_H)}{r - g_H}.$$

If reforms are *not* announced at date t , firm value satisfies

$$V_{L,t} = \frac{EC_{L,t+1} + \pi V_{LH,t+1} + (1 - \pi)V_{LL,t+1}}{1 + r}.$$

Solving recursively gives

$$V_{L,t} = EC_{L,t} \left[1 + \pi \frac{1 + g_H}{r - g_H} \right] \left[\frac{1 + g_L}{r - g_L + \pi(1 + g_L)} \right].$$

The ex post return when reforms do *not* occur is

$$R_L \equiv 1 + r_L = \frac{EC_{L,t+1} + V_{LL,t+1}}{V_{L,t}} = (1 + r) - (g_H - g_L) \left[\frac{\pi(1 + r)}{r - g_H + \pi(1 + g_H)} \right].$$

The ex post return when reforms *do* occur is

$$R_H \equiv 1 + r_H = \frac{EC_{H,t+1} + V_{HH,t+1}}{V_{L,t}} = \frac{(1 + r)(1 + g_H)}{(1 + g_L)} \cdot \frac{r + \pi - g_L(1 - \pi)}{r + \pi - g_H(1 - \pi)}.$$

Main proposition.

$$R_L < 1 + r < R_H,$$

and R_L is decreasing in π .

Interpretation.

- If investors rationally expect reforms but reforms do *not* arrive, stock returns are **below the opportunity cost of capital**.
- If reforms do arrive, there is a **jump** in prices and returns.
- The more likely investors think reform is, the lower the ex post return in no-reform years because disappointment is larger.

3.3 Calibration of the baseline model

The authors calibrate the model using:

- A-share annualized nominal return of about **4%** as a proxy for g_L ,
- externally listed Chinese firms' annualized nominal return of about **10%** as a proxy for g_H ,
- reform probabilities $\pi \in \{0.02, 0.05, 0.10\}$,
- opportunity cost values $r \in \{0.12, 0.15\}$.

With $\pi = 0.02$ and $r = 0.12$, value rises from about **21.34** to **54.48** after reform, a jump of about **155%**, while the gross return rises from about **1.09** to **2.75**, a jump of about **166 percentage points**.

Interpretation. Even infrequent reform hopes can support high valuations and low ex post returns for long periods.

3.4 Behavioral model with heterogeneous investor beliefs

There are two groups of investors:

- **optimists**, who believe reforms occur with probability π_H ,
- **realists**, who believe reforms occur with probability π ,

with $\pi_H > \pi$.

The model assumes no short sales, and total wealth satisfies

$$W_t^{OP} + W_t^{RL} > S_t.$$

If optimists are the marginal holders of the stock, low-state valuation is

$$V_{L,t}^{OP} = \frac{EC_{L,t+1} + \pi_H V_{LH,t+1} + (1 - \pi_H) V_{LL,t+1}}{1 + r} > V_{L,t}.$$

Interpretation.

- Short-sale constraints prevent realistic investors from pushing prices down.
- Optimistic investors can therefore set prices too high.
- When reforms fail to arrive, returns are especially low.
- The stronger the optimism, the lower the subsequent ex post return.

3.5 Financial repression model

The third version of the model introduces segmented capital markets and financial repression:

$$r < r^{EX},$$

where r is the domestic opportunity cost of capital and r^{EX} is the external-market opportunity cost.

Hence domestic returns satisfy

$$1 + r < 1 + r^{EX}.$$

Interpretation.

- If deposit and savings rates are artificially low in China, the opportunity cost of holding domestic stocks is also low.
- This can mechanically lower A-share returns relative to externally listed Chinese firms and other markets.

3.6 Empirical hypotheses

The model delivers three main hypotheses:

1. **Institutional deficiencies hypothesis.** Weak IPO and delisting rules and weak governance generate poor accounting performance and low stock returns.
2. **Behavioral biases hypothesis.** Investor optimism and speculative demand push prices too high and lower subsequent returns.
3. **Financial repression hypothesis.** Low domestic interest rates mechanically depress A-share returns relative to external markets.

3.7 Main empirical specifications

3.7.1 Cross-country stock return regressions

The paper estimates regressions of the form

$$R_{ict} = \alpha + \beta AShare_{ict} + \Gamma X_{c,t-1} + \Lambda Z_{i,t-1} + \delta_t + \delta_s + \varepsilon_{ict},$$

where:

- R_{ict} is annual real stock return,
- $AShare_{ict}$ is an indicator for Chinese domestic listing,
- $X_{c,t-1}$ are country-level controls,
- $Z_{i,t-1}$ are firm controls,
- δ_t and δ_s are year and industry fixed effects.

When the dependent variable is stock return, the paper generally uses **WLS** with lagged market capitalization as the weight.

Interpretation. The coefficient on the A-share indicator measures how much A-share firms underperform relative to otherwise comparable firms across countries.

3.7.2 IPO event-window regressions

For post-IPO operating performance, the paper estimates specifications like

$$\Delta Perf_i = \alpha + \beta AShare_i + \Gamma Controls_i + \delta_t + \delta_s + \varepsilon_i,$$

where $\Delta Perf_i$ is a change in:

- ROA,
- ROS,
- Total Accruals/Sales,
- or OCF/Sales

between windows such as $[-1, +1]$ and $[-2, +2]$ around the IPO year.

Interpretation. These regressions test whether firms entering the A-share market look unusually good just before IPO and then deteriorate more sharply afterward.

3.7.3 Investment and cash-flow regressions

For operating outcomes the paper estimates

$$Perf_{it} = \alpha + \beta AShare_{it} + \Gamma Controls_{i,t-1} + \delta_t + \delta_s(+\delta_{loc}) + \varepsilon_{it},$$

where $Perf_{it}$ is:

- Investment,
- Operating Cash Flows,
- or Net Cash Flows.

Interpretation. If A-share firms invest more but generate less cash flow, the problem is not only return pricing but also real operating efficiency.

3.7.4 Return-determinant regressions with governance and sentiment

The paper then estimates reduced-form return equations of the form

$$R_{it} = \alpha + \beta_1 G_{i,t-1} + \beta_2 Sent_{t-1} + \beta_3 \Delta ROA_i^{IPO} + \beta_4 ROIC_{it} + \Gamma Controls_{i,t-1} + FE + \varepsilon_{it}.$$

Interpretation.

- G captures governance quality,

- *Sent* captures investor sentiment,
- ΔROA^{IPO} captures IPO-related performance reversals,
- *ROIC* captures investment efficiency.

This is the key horse race between institutional and behavioral explanations.

3.7.5 Governance and accounting performance

Finally, the paper relates governance directly to cash-flow performance:

$$Perf_{it} = \alpha + \beta G_{i,t-1} + \Gamma Controls_{i,t-1} + FE + \varepsilon_{it}.$$

Interpretation. If governance predicts not just returns but also future cash flows, that strengthens the institutional story relative to a pure mispricing story.

4 Identification and instruments (what makes the estimates credible)

4.1 What a simple one-line explanation would miss

- **China's macro growth cannot explain the puzzle.** If fast GDP growth were enough, A-share returns should have been high. They are not.
- **Being a Chinese firm cannot explain the puzzle.** Externally listed Chinese firms are also Chinese firms exposed to the same underlying economy, yet they perform much better.
- **Low stock returns alone are not enough.** They could reflect lower discount rates or exuberant pricing.
- **Weak accounting returns alone are not enough.** They could reflect selection into listing or differences in firm composition.
- The paper's strategy is therefore to line up **stock returns, cash flows, IPO dynamics, delisting behavior, governance, and sentiment** all at once.

4.2 Main comparison margins that identify the mechanisms

- **A-share vs. listed firms from other countries.**
 - This shows that A-share underperformance is extreme even by global standards.
 - Country-level controls absorb standard macro and institutional differences.
- **A-share vs. externally listed Chinese firms.**
 - This is the most useful comparison for stock-return questions.
 - Both groups operate in China, but they face different listing environments and investor bases.
- **A-share vs. matched unlisted Chinese firms.**
 - This isolates operating performance within China.

- It shows that the listed A-share firms are not just poor relative to foreign stocks; they are also poor relative to unlisted domestic firms in net cash flows.
- **IPO event windows.**
 - Larger post-IPO drops in ROA/ROS reveal that the listing process itself is part of the problem.
 - Accrual reversals around IPO identify earnings management, especially among large A-share firms.
- **ST vs. U.S. delisting comparison.**
 - The paper shows that many very poor performers stay listed in China.
 - This is evidence on the exit side of market selection.
- **Governance and tunneling.**
 - Governance indexes and related-party outflows predict both returns and cash flows.
 - This is important because it links weak governance to **real firm outcomes**, not just price movements.
- **Sentiment and speculation.**
 - Cross-country sentiment measures, AH premia, and turnover all predict lower future returns.
 - This identifies a separate pricing channel beyond fundamentals.

4.3 Why the financial repression explanation is incomplete

- Real interest rates in China are lower than in many other emerging markets, so financial repression can help explain some low returns.
- But it **cannot** explain:
 - why externally listed Chinese firms outperform,
 - why matched unlisted Chinese firms have stronger cash flows,
 - why large A-share firms underperform small A-share firms,
 - why poor governance predicts poor accounting outcomes.
- Additional robustness checks also weaken the discount-rate story:
 - cross-listed A/H firms do not have clearly different returns from comparable A-share-only firms,
 - A-share firms do not have systematically lower betas,
 - A-share return volatility is if anything **higher**, not lower,
 - A-share valuations are not unusually depressed.

4.4 How strong is the paper's identification?

- The paper does **not** rely on a single quasi-experiment or a clean instrument.
- Its empirical design is instead a **diagnostic synthesis**:
 - multiple comparison groups,
 - multiple outcome variables,
 - theory-guided decomposition,
 - and robustness checks against standard asset-pricing alternatives.
- So the paper is strongest when read as a **structured explanation of a broad puzzle**, not as a narrow causal estimate from one exogenous shock.

5 Tables and figures

5.1 Figure 1: buy-and-hold returns across large markets and externally listed Chinese firms

- **Read.** This is the paper's most direct visual statement of the puzzle.
- The A-share market produces almost no real wealth accumulation from 2000 to 2018.
- Externally listed Chinese firms are among the best-performing groups in the figure.
- **Punchline.** The problem is **not** “China” per se. It is the **domestic listed market**.

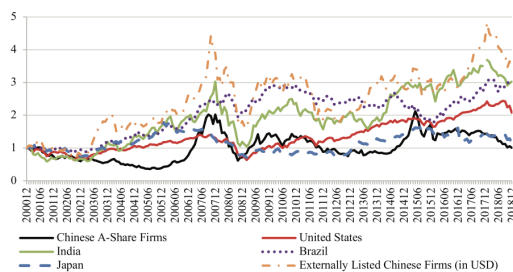


Figure 1. Buy-and-hold monthly returns of stocks listed in large countries and externally listed Chinese companies (2000 to 2018). This figure plots the value-weighted buy-and-hold returns (BHRs) of the stocks listed in the markets of China (A-share), the United States, India, Brazil, and Japan, externally listed Chinese firms, and the 30% smallest A-share firms. BHRs are calculated by cumulating value-weighted monthly returns of all stocks listed in the country. Lagged one-year market capitalization is the weight, except for externally listed Chinese firms, for which the weight is the lagged one-year market capitalization in U.S. dollars. The returns are calculated at month-end, adjusted for stock splits, and include cash dividends. Nominal returns are denominated in local currencies and adjusted for local inflation to convert to real returns. Inflation is measured by the month-average CPI rate of the listing country. The number of unique firms in the sample for China, the United States, India, Brazil, Japan, and externally listed Chinese firms is 3,695, 12,200, 4,819, 535, 4,648, and 1,770, respectively. (Color figure can be viewed at wileyonlinelibrary.com)

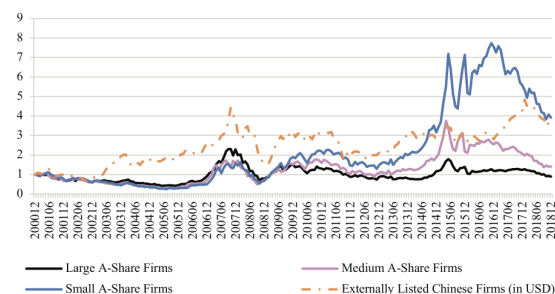


Figure 2. Buy-and-hold monthly returns of large, medium, and small-cap stocks listed in the A-share market. This figure plots the value-weighted average buy-and-hold returns BHRs of small, medium, and large A-share firms and externally listed Chinese firms, with lagged one-year market capitalization as the weight. Small and large firms are determined by the 30th and 70th percentiles of lagged one-year market capitalization by year. (Color figure can be viewed at wileyonlinelibrary.com)

5.2 Figure 2: buy-and-hold returns by A-share firm size

- **Read.** Split the A-share market into small, medium, and large firms.
- The underperformance is concentrated among **large-cap A-share firms**.
- Small-cap A-share firms perform much better, especially in the second half of the sample.

- **Punchline.** The average failure of the A-share market is not uniform across firms.

5.3 Table II: stock returns of A-share firms

- **Read.** This table quantifies the stock-return gap in regressions with country and firm controls.
- In the cross-country sample, A-share firms underperform by roughly **14.6% to 16.0% per year**.
- Externally listed Chinese firms do **not** underperform; the gap between A-shares and external Chinese firms is statistically meaningful.
- Excluding IPO years still leaves a large negative A-share coefficient.
- The smallest 30% of A-share firms do not underperform; the largest 30% underperform by about **15.5 percentage points** annually.
- **Punchline.** Low A-share returns are large, persistent, and driven by big firms rather than by the IPO window alone.

Table II
Underperformance of the A-Share Market: Stock Returns

This table examines the stock performance of A-share firms, firms listed in other countries, and externally listed Chinese firms. The dependent variable is annual stock returns in percentage points, including cash dividends and adjusted for stock splits and inflation. For firms that are delisted from an exchange, we set the delisting return (stock return in the year after the firm is delisted) to -30%. In Panel A, the independent variable of interest is *A-share Listed*, a dummy variable that takes the value of one if the stock is listed in the SSE or SZSE, and zero otherwise. The coefficients on *GDP Growth*, *Market Turnover*, *EBIT of Listed Firms/GDP*, and *Consumption Volatility* are multiplied by 100. In Panel B, the dummy variable of interest is *Externally Listed Chinese Firms*. In Panel C, we exclude returns in the IPO year (year 0) and year +1 from the regression sample. All control variables are lagged one year. Exchanges that have fewer than 20 stocks in any given year are excluded from the sample. We control for year and industry fixed effects based on the Datastream level-2 industry classifications. Panel D reports cross-country regressions of stock returns: column (1) includes all A-share firms, column (2) excludes A-share SOEs, column (3) excludes the largest 30% A-share firms by lagged one-year market capitalization, and column (4) (5) focuses on the smallest (largest) 30% A-share firms only. We employ WLS regressions where we use lagged one-year market capitalization as the weight. *t*-Statistics calculated using standard errors clustered by industry and year are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Panel A. The Cross-Country Sample

	1991 to 2018		2000 to 2018	
	(1)	(2)	(3)	(4)
A-Share Listed	-14.588*** (-6.044)	-13.399*** (-4.732)	-15.969*** (-5.291)	-14.948** (-2.389)
GDP Growth	1.544*** (6.533)	0.626* (1.744)	0.465 (1.399)	0.535 (1.322)
Log (GDP Per Capita)	-2.852*** (-2.598)	-1.402 (-1.422)	-1.968** (-2.119)	-1.967* (-1.966)
M2/GDP	1.809 (0.963)	4.744** (2.375)	1.663 (0.822)	5.848** (2.509)
Market Turnover	6.408** (2.311)	3.059 (1.402)	1.091 (0.522)	2.518 (1.106)
Credit from Financial Institutions/GDP	-1.879 (-1.033)	-6.711** (-2.093)	0.528 (0.168)	-8.271** (-2.344)
EBIT of Listed Firms/GDP	0.008 (-0.552)	0.007 (0.625)	-0.029* (-1.873)	-0.032* (-1.809)
Consumption Volatility	-2.003* (-1.833)	-0.540 (-1.435)	0.386 (0.445)	1.253 (1.277)
Anti-self-dealing Index	-2.498 (-0.633)	7.710** (2.062)	5.494* (1.653)	8.172** (2.099)
Tax Evasion	-0.603 (-0.611)	-0.812 (-0.822)	-1.448 (-1.392)	-1.170 (-1.052)
Time to Collect on a Bounced Check	1.354 (1.286)	3.583*** (2.766)	2.933** (1.969)	2.781* (1.922)
Log (Total Assets)		0.006 (0.029)		-0.410 (-1.577)
Leverage		5.381 (1.263)	8.209* (1.752)	

(Continued)

Table III
Correlations between (Five-Year) Stock Returns and Future GDP Growth Rates

This table reports Pearson correlations between stock returns and future GDP growth rates for the 20 largest economies according to the 2018 IMF rankings of GDP in PPP terms. We also include South Africa as a large emerging economy. We estimate the correlation coefficients for the period 1991 to 2018, or for a period starting from the year (after 1991) in which stock returns data become available in our data set and ending in 2018. Correlations are estimated using cumulative stock returns of a five-year interval and cumulative GDP growth in the next five-year interval (so we obtain stock returns for years t to $t+4$, $t+1$ to $t+5$, and so forth, and GDP growth for years $t+1$ to $t+5$, $t+2$ to $t+6$, ...), until the end of 2018, on a rolling basis. Country-level stock returns are calculated as the value-weighted returns of individual stocks listed in a country, with lagged one-year market capitalization as the weight. For South Korea and Iran, we estimate the correlation coefficients using the stock market indices (KOSPI Korea and TEPIX Iran). We calculate value-weighted returns for externally listed Chinese firms and estimate their correlations using China's GDP growth rate in year $t+1$. The last three rows present results on the differences in the correlation coefficients of China A-share firms versus developed countries as a group, other emerging countries as a group, and externally listed Chinese firms as a group. We use the OECD classification to define developed and emerging countries; emerging economies include China, Brazil, the Russian Federation, India, Mexico, Indonesia, Turkey, Iran, Thailand, Saudi Arabia, and South Africa.

IMF GDP (PPP) Ranking	Country	Sample Period	Correlation	p-Value
1	China A-Share	1991 to 2018	-0.013	0.637
2	United States	1991 to 2018	0.286***	0.001
3	India	1991 to 2018	0.185***	0.001
4	Japan	1991 to 2018	0.380*	0.072
5	Germany	1991 to 2018	0.532***	0.008
6	Russian Federation	1996 to 2018	0.386***	0.001
7	Indonesia	1991 to 2018	0.531**	0.011
8	Brazil	1995 to 2018	0.429***	0.001
9	United Kingdom	1991 to 2018	0.479**	0.029
10	France	1991 to 2018	0.587***	0.004
11	Italy	1991 to 2018	0.446*	0.072
12	Mexico	1991 to 2018	0.489**	0.013
13	Turkey	1991 to 2018	0.195	0.396
14	South Korea	1991 to 2018	0.366	0.147
15	Spain	1991 to 2018	0.600***	0.0032
16	Saudi Arabia	1995 to 2018	0.350**	0.035
17	Canada	1991 to 2018	0.521***	0.009
18	Iran	1997 to 2018	0.056	0.280
19	Thailand	1991 to 2018	0.469***	0.001
20	Australia	1991 to 2018	0.116	0.656
*	South Africa	1991 to 2018	0.656***	0.001
Externally Listed Chinese Firms			1998 to 2018	0.410**
HK Listed Chinese Firms			1998 to 2018	0.555**
Difference	Average Correlation	China	Other Groups - China	p-Value
Developed	0.437***	-0.013	0.450***	0.001
Emerging	0.375***	-0.013	0.388***	0.001
Externally Listed Chinese Firms	0.410**	-0.013	0.423**	0.015

5.4 Table III: stock returns and future GDP growth

- **Read.** This table asks whether the stock market predicts future macro performance.
- For China A-shares, the correlation between five-year stock returns and future five-year GDP growth is about **-0.013** and insignificant.

- For externally listed Chinese firms, the same correlation is about **0.410** and significant.
- **Punchline.** The domestic stock market is disconnected from the broader Chinese growth process.

5.5 Table IV: model calibration

- **Read.** This table maps the theoretical model into plausible magnitudes.
- With modest reform probabilities and reasonable opportunity costs, the model generates large valuation jumps and large return jumps after reform.
- **Punchline.** Expectations of reform can sustain long periods of low realized returns without requiring obviously irrational investors.

Table IV
Model Calibration (Version with Common Beliefs)

This table presents calibration results based on the first version of the model (with common investor beliefs on the likelihood of institutional and governance reforms) and Proposition 1. The probability of implementing reforms, π , takes three values: 0.02, 0.05, and 0.1. EC_t is mean cash flows when institutions and corporate governance are in a low state. We normalize EC_t to be one. g_L is the growth rate of cash flows per period when institutions and governance are in a low state. g_H is the growth rate of cash flows between t and $t+1$ when reforms are announced and implemented in t . We use annualized nominal stock returns of A-share firms and externally listed Chinese firms during 2000 to 2018 to proxy for g_L and g_H . r is the opportunity cost of capital for funding the representative firm's risky project. V_{LH} and V_L represent firm value before and after the implementation of reforms, and $R_H - R_L$ represents the jump in returns that investors realize in the current period when reforms are announced and implemented.

Model Input	$r = 0.12$			$r = 0.15$		
	(1)	(2)	(3)	(4)	(5)	(6)
π	0.02	0.05	0.10	0.02	0.05	0.10
EC_t	1.00	1.00	1.00	1.00	1.00	1.00
g_H	0.10	0.09	0.09	0.10	0.09	0.09
g_L	0.04	0.04	0.04	0.04	0.04	0.04
R	0.12	0.12	0.12	0.15	0.15	0.15
Derived Valuation						
V_H	52.33	34.78	34.78	21.55	17.76	17.76
V_{LH+1}	54.48	36.21	36.21	22.43	18.49	18.49
V_L	21.34	21.76	25.46	11.47	12.20	13.56
Stock Return						
R_H	2.75	1.79	1.53	2.16	1.67	1.51
R_L	1.09	1.09	1.08	1.13	1.13	1.12
$R_H - R_L$ (in percentage points)	166%	70%	45%	103%	55%	39%
$V_{LH} - V_L$ (in %)	155%	66%	42%	95%	51%	36%

Table V

Operating Performance around IPO: China versus Other Countries

This table presents OLS regression results for changes in ROA and ROS around IPO for listed firms in our cross-country sample. We calculate the changes in ROA and ROS from year $t-1$ to year $t+1$, and from year $t-2$ to year $t+2$; year t represents the IPO year. Panel A reports results of the sample of A-share firms and firms listed in other countries. Panel B reports results of the sample of A-share firms and externally listed Chinese firms. The independent variable of interest is the *A-share Listed* indicator. We control for industry-median ROA or ROS measured in year $t-1$ or $t-2$, and the change in cash holdings from year $t-1$ to t , scaled by book assets in year $t-1$. t -Statistics calculated from standard errors clustered by industry and year are reported in parentheses. ***, **, and * denote the statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variable definitions.

Panel A. A-Share Listed Firms versus Firms Listed in Other Countries

	$\Delta ROA [-1, +1]$ (1)	$\Delta ROS [-1, +1]$ (2)	$\Delta ROA [-2, +2]$ (3)	$\Delta ROS [-2, +2]$ (4)
A-Share Listed	-0.024*** (-3.022)	-0.021** (-2.069)	-0.040*** (-7.656)	-0.036*** (-2.983)
ROA ($t-1$) or ($t-2$)	-0.223*** (-2.996)		-0.364*** (-9.191)	
ROS ($t-1$) or ($t-2$)		-0.255*** (-3.102)		-0.241*** (-19.733)
Log (Total Assets)	0.005*** (8.132)	0.003** (2.203)	0.002** (2.181)	0.001 (0.005)
Sales Growth	-0.028*** (-8.647)		-0.016* (-1.932)	
EBIT Growth		-0.015*** (-5.970)		-0.022*** (-13.355)
Δ Cash Holdings $[-1, 0]$	-0.002 (-0.100)	0.055*** (2.622)	0.054 (1.255)	0.114 (1.628)
Leverage	0.084*** (8.990)	-0.001 (-0.143)	0.112*** (7.282)	-0.025 (-1.281)
Log (1+Firm Age)	0.008*** (5.722)	0.004 (0.860)	0.009*** (3.291)	0.009 (1.267)
GDP Growth	-0.099 (-1.269)	-0.066 (-0.417)	-0.067 (-0.709)	-0.142 (-1.055)
Log (GDP Per Capita)	-0.001 (-0.145)	0.008*** (2.623)	0.001 (0.541)	0.005 (1.103)
Year/Ind FE	Yes	Yes	Yes	Yes
R^2 (%)	5.33	1.59	4.39	2.82
Observations	35,604	28,022	26,603	20,544

Panel B. A-share Listed Firms vs. Externally Listed Chinese Firms

	$\Delta ROA [-1, +1]$ (1)	$\Delta ROS [-1, +1]$ (2)	$\Delta ROA [-2, +2]$ (3)	$\Delta ROS [-2, +2]$ (4)
A-Share Listed	-0.021* (-1.649)	-0.018*** (-2.869)	-0.030** (-1.982)	-0.025** (-1.990)
Year/Industry/Firm Location FEs	Yes	Yes	Yes	Yes
Other Firm Controls	Yes	Yes	Yes	Yes
R^2 (%)	55.21	31.43	37.63	22.16
Observations	4,173	4,173	2,398	2,398

5.6 Table V and Table VI: IPO dynamics and earnings management

- **Read.** These tables test whether the listing process creates adverse selection.
- Relative to firms in other countries, A-share firms experience larger declines in:
 - **ROA:** about **-2.4 percentage points** over the $[-1, +1]$ IPO window,
 - **ROS:** about **-2.1 percentage points** over the same window,
 - with even larger drops over the $[-2, +2]$ window.

- Relative to externally listed Chinese firms, A-share firms also exhibit sharper post-IPO operating deterioration.
- Large A-share firms show a much larger post-IPO fall in **accruals**, consistent with earnings management before listing.
- **Punchline.** The IPO process in China appears to reward firms that can temporarily look profitable enough to clear the listing hurdle.

Table VI
Earnings Management around IPO: China versus Other Countries

This table reports OLS regression results for changes in earnings management measures in the $[-1, +1]$ window around IPO for A-share firms and firms listed in other countries, where year 0 denotes the IPO year. Panels A and B compare *Total Accruals/Sales* and *OCF/Sales* of A-share firms and firms listed in other countries. We construct *Total Accruals* as Net Income – Operating Cash Flows, *Operating Cash Flows (OCF)* are calculated as EBITDA – Income Taxes – Change in Working Capital. We use the same country- and firm-level controls as we use in Table V. ***, ** and * denote the statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variable definitions.

Panel A. Cross-Country Sample: Total Accruals/Sales				
	All (1)	Large (2)	Medium (3)	Small (4)
A-Share Listed	-0.025 (-0.789)	-0.103** (-2.010)	-0.078 (-0.944)	-0.042 (-0.382)
Year/Ind FE	Yes	Yes	Yes	Yes
Country Controls	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
R ² (%)	6.78	11.78	9.65	9.88
Observations	18,757	8,657	3,141	6,959

Panel B. Cross-Country Sample: OCF/Sales				
	All (1)	Large (2)	Medium (3)	Small (4)
A-Share Listed	0.043 (1.038)	0.023 (0.536)	0.083 (0.891)	-0.010 (-0.398)
Year/Ind FE	Yes	Yes	Yes	Yes
Country Controls	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
R ² (%)	7.05	6.83	8.82	8.52
Observations	19,040	8,826	3,169	7,045

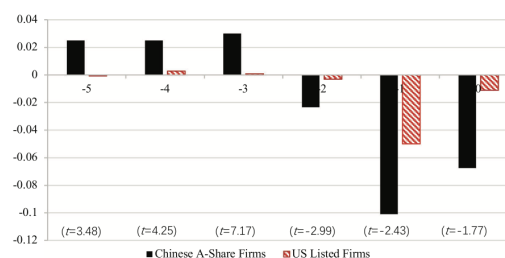


Figure 4. Operating performance of “ST” firms in China and delisted firms in the United States. This figure plots the operating performance of permanent ST A-share firms in the $[-5, 0]$ window before receiving the special treatment (ST) status for the first time, and the operating performance of delisted U.S. firms in the same event window. Year 0 denotes the year when a firm receives ST status (A-share firms) or the year a U.S. firm is delisted. Operating performance is measured as the weighted averaged ROA across firms, with lagged one-year total assets as the weight. t -Statistics for the difference in ROA between A-share and U.S. firms are reported for each year. (Color figure can be viewed at wileyonlinelibrary.com)

5.7 Figure 4: ST firms in China versus delisted U.S. firms

- **Read.** This figure compares firms that would likely exit stricter markets.
- Chinese permanent ST firms deteriorate sharply and are worse than U.S. delisted firms in the years immediately before the event.
- **Punchline.** The A-share market keeps poor-performing firms alive instead of removing them, which worsens market composition.

5.8 Figure 5 and Table VII: investment and cash flows

- **Read.** These are the core accounting-performance exhibits.
- A-share firms invest more than global peers but generate less net cash flow.
- In cross-country regressions, A-share firms invest about **0.551 percentage points** more and generate about **0.766 percentage points** less net cash flow per year.
- Relative to matched unlisted Chinese firms, A-share firms invest about **2.22 percentage points** more, generate about **4.07 percentage points** lower OCF, and about **3.63 percentage points** lower net cash flow.

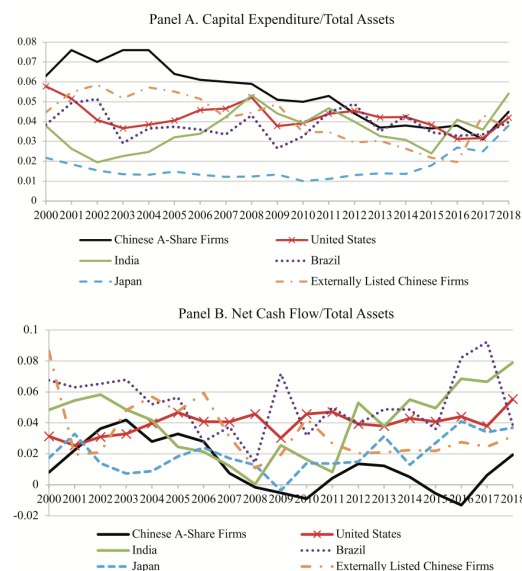


Figure 5. Investment and net cash flows of A-share firms, externally listed Chinese firms, and firms from other large countries. This figure plots the weighted-average investment and net cash flows of A-share firms, firms listed in other large countries, and externally listed Chinese firms by calendar year. Panel A plots the weighted-average investment of listed firms, where *Investment* is measured as Capital Expenditures/Total Assets. Panel B plots the weighted-average net cash flows of listed firms, where *Net Cash Flow* is calculated as (EBITDA – Change in Working Capital – Income Taxes – Capital Expenditures)/Total Assets. [Both the investment and net cash flow measures are value-weighted averages across firms with year-end total assets as weights.] The sample is restricted to firms that have nonmissing data on investment, net cash flows, and book assets. The number of unique firms in the sample for China, the United States, India, Brazil, Japan, and externally listed Chinese firms is 3,695, 10,529, 3,708, 321, 4,183, and 1,487, respectively. (Color figure can be viewed at wileyonlinelibrary.com)

- Among large firms, A-share firms underperform matched externally listed Chinese firms in net cash flows.
- **Punchline.** The problem is not just low stock returns; listed domestic firms also convert investment into cash poorly.

5.9 Table VIII, Table IX, Table X, and Table XI: governance and sentiment

- **Read.** These tables compare the institutional and behavioral explanations directly.
- The cross-country governance index **G-Index-I** is lower for A-share firms than for Hong Kong and U.S. firms.
- Within China, the A-share governance index **G-Index-A** is higher for **small firms** than for large firms, and higher for **non-SOEs** than for SOEs.
- In return regressions:
 - stronger governance predicts **higher future returns**,
 - higher sentiment predicts **lower future returns**.
- For example, in the cross-country regressions a one-standard-deviation rise in sentiment reduces returns by roughly **4.4 to 6.8 percentage points**, while governance raises returns.
- Within the A-share sample, better governance raises returns, while higher turnover lowers them.
- Governance also predicts better **operating cash flows** and **net cash flows**.

Table VII
Comparing Investment and Cash Flows of A-Share Firms with Other Groups of Firms

This table reports OLS regression estimates from comparing investment and net cash flows of A-share firms with other groups of firms. The dependent variables are *Investment*, measured as Capital Expenditures/Total Assets, *Operating Cash Flows*, calculated as EBITDA - Change in Working Capital - Income Taxes/Total Assets, and *Net Cash Flows*, calculated as (Operating Cash Flows - Capital Expenditures)/Total Assets. All dependent variables are reported in percentage points. The independent variable of interest is *A-Share Listed*, a dummy variable that takes a value of one if the firm is listed in the Shanghai or Shenzhen stock exchanges in mainland China, and zero otherwise. Panel A reports estimates from comparing A-share firms and listed firms from other countries in our sample. Firm controls are *Lag (Total Assets)*, *Leverage*, *ROA*, *Sales Growth*, and *Lag (Firm Age)*. Control variables are lagged one year. Panel B reports cross-country regressions of net cash flows, column (1) includes all A-share firms, column (2) excludes A-share SOEs, column (3) excludes the largest 30% A-share firms by lagged one-year market capitalization, and column (4) (5) studies the smallest (largest) 30% A-share firms only. In Panels A and B, we include *GDP Growth* and *GDP Per Capita* as country-level controls. Panel C reports estimates from comparing A-share firms and size- and industry-matched unlisted Chinese firms. We employ nearest-neighbor matching and require the ratio of the total assets of the A-share firm and to its matched unlisted firm be within the [80%, 120%] range. Panel D reports estimates from comparing A-share firms and size-sizes- and industry-matched externally listed Chinese firms. We follow the same matching procedure as in Panel C. In Panels D1 to D3, we divide the matched sample of A-share firms and externally listed Chinese firms into small, medium, and large groups based on the 30th and 70th percentiles of lagged one-year market capitalization of A-share firms. We control for lagged one-year firm characteristics and include year and industry fixed effects in all specifications. In Panels C and D, we also include province-level firm location fixed effects. *t*-Statistics calculated using standard errors clustered by industry and year are reported in parentheses. ***, **, and * denote the statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variable definitions.

Panel A. Comparing A-Share Firms with Firms from Other Countries			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flows (3)
A-Share Listed	0.551*** (4.353)	-0.221 (-0.733)	-0.769*** (-2.645)
Lag (Total Assets)	-0.188*** (8.128)	0.867*** (16.720)	1.088*** (18.125)
Leverage	0.207 (0.386)	4.014*** (6.110)	4.123*** (5.489)
Lag (1+ Firm Age)	-0.423*** (-10.004)	0.345*** (4.028)	0.345*** (7.384)
Sales Growth	-0.037*** (6.548)	-0.037*** (-2.201)	-0.037*** (-4.132)
ROA	1.121*** (44.173)	56.26*** (44.921)	54.981*** (44.033)
Year/Industry FE	Yes	Yes	Yes
Country Controls	Yes	Yes	Yes
R ² (%)	11.27	28.18	26.79
Observations	546,946	546,798	546,798

(Continued)

Table VIII
Summary Statistics for Corporate Governance Measures

This table presents summary statistics to the governance index for A-share firms and for other markets, as well as for the components of the G-Index. Panel A reports results for the governance measure, G-Index-I, for the cross-country sample. The G-Index-I ranges from zero to seven. It is the sum of the scores on the following four dimensions: CEO duality, board size, ownership concentration, and executive holdings. The score for CEO duality takes a value of one if the firm's CEO and chairman is the same person, and zero otherwise. The score on board size ranges from zero to two: it takes a value of two if the firm's board size (measured by the number of directors) is smaller than the 30th percentile, one if between the 30th and 70th percentiles, and zero if larger than the 70th percentile. The scores on ownership concentration and executive holdings both range from zero to two: they take a value of two if the variable is larger than the 70th percentile, one if between the 30th and 70th percentiles, and zero if smaller than the 30th percentile. Percentile points are determined by country and year. We report the value-weighted average G-Index-I for each group, with lagged one-year market capitalization as the weight. The G-Index-I is available for firms listed in 75 countries in our sample. Panel B reports summary statistics for the governance measure for A-share firms, G-Index-A, and its components. G-Index-A ranges from zero to six and is the sum of scores on the following six dimensions: CEO duality, state ownership, executive holdings, board size, ownership concentration, and related-party transactions (RPT). We add one to the G-Index-A for each of the following conditions if met: (i) the firm's CEO and chairman is the same person, (ii) the firm's state holdings (central SASAC, local SASAC, and other government agencies) are less than 40%, (iii) the proportion of shares held by the firm's executives exceeds the 70th percentile of the year, (iv) board size is below the 30th percentile, (v) the proportion of shares held by the firm's largest five shareholders exceeds the 70th percentile, and (vi) the firm does not have net cash outflows in RPT in the year. In the bottom of Panel B, we compare the average G-Index-A for large and small A-share firms and for SOEs and non-SOEs. ***, **, and * denote the statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variable definitions.

Panel A. Governance Measure and Its Components for the Cross-Country Sample (G-Index-I)						
Country	Component	Mean	Median	StDev	Max	Min
Chinese A-share	CEO Duality	0.398	0	0.496	1	0
	Ownership Concentration	0.539	0.546	0.152	0.891	0.201
	Executive Holdings	0.104	0.001	0.196	0.681	0
	Board Size	16.8	16	4.029	54	1
	G-Index-I	2.271	2	1.527	7	0
Hong Kong	CEO Duality	0.022	0	0.146	1	0
	Ownership Concentration	0.312	0.250	0.321	1	0
	Executive Holdings	0.277	0.147	0.286	0.717	0.0001
	Board Size	9.7	9	2.838	22	4
	G-Index-I	3.069	3	1.150	7	0
United States	CEO Duality	0.090	0	0.286	1	0
	Ownership Concentration	0.066	0.057	0.068	1	0
	Executive Holdings	0.102	0.001	0.443	1	0
	Board Size	9.7	9	2.373	33	1
	G-Index-I	2.406	2	0.862	7	0
Other Countries	CEO Duality	0.398	0	0.496	1	0
	Ownership Concentration	0.116	0	0.211	1	0
	Executive Holdings	0.458	0.478	0.262	0.923	0
	Board Size	8.7	8	4.452	34	1
	G-Index-I	2.365	2	0.788	7	0

(Continued)

Table VII—Continued

Panel B. Net Cash Flows of A-Share Firms: Alternative Samples					
	Full Sample (1)	Drop SOEs (2)	Drop Large (3)	Small Only (4)	Large Only (5)
A-Share Listed	-0.766*** (-2.645)	-1.110*** (-2.833)	-1.148* (-1.762)	-0.252 (-0.277)	-2.052** (-3.398)
Country/Firm Controls	Yes	Yes	Yes	Yes	Yes
Year and Ind FE	Yes	Yes	Yes	Yes	Yes
R ² (%)	26.79	26.62	30.07	30.21	30.44
Observations	546,798	528,617	533,236	516,620	512,926

Panel C. A-Share Firms vs. Matched Unlisted Chinese Firms			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flow (3)
A-Share Listed	2.220* (1.815)	-4.070** (-2.253)	-3.633** (-1.990)
Year/Ind/Location FEs	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes
R ² (%)	11.77	13.23	6.18
Observations	19,218	18,639	18,638

Panel D. A-Share Firms vs. Matched Externally Listed Chinese Firms			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flow (3)
A-Share Listed	0.0402 (0.311)	0.391 (1.325)	0.382 (1.366)
Year/Ind/Location FEs	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes
R ² (%)	19.20	11.10	4.43
Observations	17,886	17,353	17,353

Panel D1. Large A-Share Firm and Matched Externally Listed Chinese Firms			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flow (3)
A-Share Listed	0.404** (2.198)	-0.072 (-0.029)	-0.653** (-1.960)
Year/Ind/Location FEs	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes
R ² (%)	43.32	27.93	11.07
Observations	2,546	2,316	2,316

Panel D2. Medium A-Share Firm and Matched Externally Listed Chinese Firms			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flow (3)
A-Share Listed	0.504** (2.225)	0.597 (1.152)	-0.193 (-0.402)
<i>(Continued)</i>			

(Continued)

Table VII—Continued

Panel D2. Medium A-Share Firm and Matched Externally Listed Chinese Firms			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flows (3)
Year/Ind/Location FEs	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes
R ²	20.54	11.54	5.03
Observations	5,073	4,970	4,970

Panel D3. Small A-Share Firm and Matched Externally Listed Chinese Firms			
	Investment (1)	Operating Cash Flows (2)	Net Cash Flows (3)
A-Share Listed	-0.684*** (-3.375)	0.374 (0.633)	0.962* (1.709)
Year/Ind/Location FEs	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes
R ²	15.32	10.52	4.64
Observations	9,834	9,639	9,639

Table IX
Summary Statistics for Investors' Behavioral Biases Measures

Panel A presents summary statistics for the two (market-level) sentiment measures and their components. *Sentiment₁* is estimated as the first principal component of volatility premium (PVOL) and market turnover (TURN), which are available for 58 countries in our sample. *Sentiment₂* is estimated as the first principal component of *L* of IPOs (*L* of IPOs), IPO first-day returns (RFP), PVOL, and TURN, which are all available for 17 countries. We report the value-weighted average *Sentiment₁* and *Sentiment₂* for each group of firms, with lagged one-year market capitalization as the weight. Panel B reports summary statistics for the investor behavior measures for A-share firms: the annualized market-level AH premium and individual stock-level turnover. We calculate *AH Premium* as the ratio of the A-share price in USD to the H-share price in USD/100 using stocks cross-listed in the A- and H-share markets. To obtain the market-level measure, we take value-weighted average across firms with the total market value of each pair A and H shares as weights. Annual stock-level turnover is estimated as annual trading volume divided by the number of shares. Appendix A provides detailed variable definitions.

Panel A. Summary Statistics of Sentiment Variables and Their Components						
Country	Component	Mean	Median	StDev	Max	Min
Chinese A-share	PVOL	0.385	0.305	0.253	0.937	-0.002
	L of IPO	4.773	4.732	0.774	6.078	2.708
	RFP	1.366	1.149	1.170	4.235	0.000
	TURN	0.208	0.135	0.358	0.959	-0.353
	Sentiment ₁	0.723	0.322	0.966	1.688	-1.242
Hong Kong	PVOL	0.477	0.436	0.367	0.921	0.011
	L of IPO	3.823	4.094	1.112	5.975	0.000
	RFP	0.389	0.107	0.735	2.847	0.007
	TURN	0.114	-0.084	0.728	2.585	-0.498
	Sentiment ₁	0.456	0.042	1.121	2.412	-2.353
United States	PVOL	0.574	0.568	0.273	1.110	-0.248
	L of IPO	4.544	4.673	0.639	5.940	3.045
	RFP	0.157	0.139	0.106	0.564	0.057
	TURN	0.028	-0.037	0.416	1.024	-0.601
	Sentiment ₁	-0.311	-0.442	0.916	1.988	-1.650
Other Countries	PVOL	0.123	0.197	0.469	1.144	-1.238
	L of IPO	3.101	3.113	1.105	5.313	0.000
	RFP	0.358	0.182	0.295	1.433	-0.094
	TURN	-0.004	-0.038	0.272	1.200	-0.758
	Sentiment ₁	0.260	0.056	1.068	3.366	-3.102
Panel B. Summary Statistics of Investor Behavior Measures for the A-Share Sample						
	Mean	Median	StDev	Min	Max	
Market-Level AH Premium	1.028	0.843	1.33	0.062	6.604	
Stock-Level Turnover	3.148	2.487	2.383	0.071	12.178	

- **Punchline.** Behavioral forces matter for returns, but governance matters for **both** returns and fundamentals.

Table X
Comparisons of Return Factors: Institutional Features, Corporate Governance, and Investors' Behavioral Biases

This table compares the effects of corporate governance, investor behavior, and earnings management, among other factors, on annual stock returns. Panel A presents results for the cross-country sample; Panel B presents the results for the A-share sample. In both panels, the dependent variable is annual stock returns in real terms. In Panel A, the explanatory variables are the global governance measure, *G-Index-I*, the investor sentiment measures *Sentiment_t* and *Sentiment_{t-1}*, $\Delta ROA[-1, +1]$, the change in ROA from the year prior to IPO to the year after, and contemporaneous investment efficiency, *ROIC*. In Panel B, the explanatory variables are the A-share governance measure *G-Index-A* and two investor behavioral measure, namely, annual stock-level turnover and the industry-level AH premium, both lagged one year. *Stock-Level Turnover* is annual trading volume divided by the number of shares. *AH Premium* is the ratio of the A-share price in USD to the H-share price in USD*100. To obtain the industry-level measure, we take value-weighted average across firms by industry, with the total market value of each pair's A and H shares as the weight. *Firm controls* include firm characteristics, province-level location fixed effects, and SOE and ST indicators. All continuous explanatory variables are normalized by dividing the original value by its standard deviation. We employ WLS regressions where we use lagged one-year market capitalization as the weight. *t*-Statistics calculated using standard errors clustered by industry and year are reported in parentheses. Appendix A provides detailed variable definitions.

Panel A. Cross-Country Sample								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
G-Index-I	1.407** (2.414)				1.330** (2.291)	1.138** (2.225)	1.411** (2.425)	1.632*** (2.826)
Sentiment _t		-4.673*** (-4.385)			-4.402*** (-2.892)			
Sentiment _{t-1}			-8.862*** (-6.471)			-7.091*** (-3.382)		-6.775*** (-3.316)
ΔROA [-1, +1]				1.973*** (2.784)			2.076** (2.473)	2.061** (2.455)
ROIC								5.714*** (6.393)
A-share Listed	-14.789** (-2.355)	-29.824*** (-3.052)	-2.277 (-0.422)	-23.318** (-2.289)	-15.580** (-2.214)	-9.566 (-1.182)	-14.627** (-2.337)	-3.306 (-0.412)
Year/Ind. FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ² (%)	5.33	6.32	6.72	5.30	6.27	6.75	5.34	7.34
Observations	495,651	423,418	411,885	495,651	424,940	411,885	495,651	411,885

Panel B. A-Share Listed Firms								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
G-Index-A	1.445*** (2.885)				1.358*** (2.556)	1.359*** (2.795)	1.341** (2.355)	0.914* (1.906)
AH Premium (Industry level)		-14.024** (-2.335)			-0.951 (-0.304)			
Stock-Level Turnover			-2.475*** (-3.866)			-1.617*** (-2.177)	-1.568* (-1.892)	-0.720 (-0.902)
Δ ROA [-1, +1]				0.869** (2.115)			0.863** (2.093)	0.051 (0.133)
ROIC								8.000*** (8.013)
Ind./Location FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
Country Controls	No	Yes	No	No	No	No	No	No
R ² (%)	52.40	4.27	52.33	52.41	53.67	53.27	53.29	54.40
Observations	36,044	28,251	34,386	36,017	28,251	34,406	34,406	31,712

Table XI
Corporate Governance and Operating Performance

This table reports results on the role of corporate governance in explaining firms' operating performance. Panel A reports results for the cross-country sample. The dependent variable is annual operating performance as measured by investment, OCF, and net cash flows. The independent variable of interest is *G-Index-I*. Firm controls are the same as in Table VII, Panel A. Both *G-Index-I* and firm controls are lagged one year. Panel B reruns the analysis of Panel A after excluding the largest 30% A-share firms from the cross-country sample. Panel C examines the relationship between governance and operating performance for A-share firms. Firm controls include *Log (Market Capitalization)* and *Market-to-Book Ratio*. Columns (1) to (3) employ the full sample period, 2000 to 2018, whereas column (4) employs the subsample period 2010 to 2018. The dependent variables are the same operating performance measures as above. The independent variable of interest is the governance measure for A-share firms, *G-Index-A*. *G-Index-A* is constructed at firm-year level. Both *G-Index-A* and controls are lagged one year. *t*-Statistics calculated using standard errors clustered by industry and year are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variable definitions.

Panel A. Governance and Operating Performance in Cross-Country Sample			
	Investment (1)	OCF (2)	Net Cash Flows (3)
G-Index-I	0.136*** (3.492)	0.893*** (10.352)	0.723*** (6.145)
A-Share Listed	0.393** (2.075)	-0.325*** (-2.683)	-0.705** (-2.465)
Country/Firm Controls	Yes	Yes	Yes
Ind/Year FE	Yes	Yes	Yes
R ² (%)	11.07	35.58	34.13
Observations	454,499	454,499	454,499

Panel B. Governance and Operating Performance in Cross-Country Sample: Dropping Large A-Share Firms			
	Investment (1)	OCF (2)	Net Cash Flows (3)
G-Index-I	0.134*** (3.223)	0.943*** (10.429)	0.771*** (6.195)
A-Share Listed	0.101 (0.513)	-0.199 (-0.542)	-0.073 (-0.162)
Country/Firm Controls	Yes	Yes	Yes
Ind/Year FE	Yes	Yes	Yes
R ² (%)	11.09	35.56	34.13
Observations	443,505	443,505	443,505

Panel C. Governance and Operating Performance in A-Share Sample				
	Investment (1)	OCF (2)	Net Cash Flows (3)	Net Cash Flows (2010-2018) (4)
G-Index-A	0.387*** (11.452)	0.448*** (4.812)	0.066 (0.823)	0.136* (1.782)
Year/Ind/Province FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
R ² (%)	12.52	10.23	2.32	2.31
Observations	31,085	29,885	29,885	19,153

5.10 Table XII and Table XIII: heterogeneity and robustness

- **Read.** These tables show where the underperformance comes from and what alternative explanations fail.
- Large A-share firms underperform other A-share firms in both stock and accounting returns.
- Decomposition shows that among all A-share firms:
 - **incumbent firms** account for about **48.7%** of the stock-return gap,
 - **newly listed firms** account for about **30.9%**,
 - **ST firms** account for about **20.4%**.
- Cross-listed A/H firms do not exhibit obviously stronger returns or cash flows than comparable A-share firms.
- **Punchline.** The underperformance is broad-based within the domestic market and is not explained away by discount-rate or cross-listing stories.

Table XII
Summary of Additional Tests and Robustness Checks

This table reports results on heterogeneity in the cross-country sample and in the A-share sample in terms of stock returns and operating performance. Panel A compares the underperformance of the largest 30% A-share firms and SOEs within the A-share firm sample. Panel B decomposes the performance gaps of A-share firm across (i) newly listed firms (firms that conduct IPO within the past two years), (ii) incumbent firms, and (iii) ST firms across three groups of A-share firms, namely, all A-share firms, SOEs, and non-SOEs. We first allow all A-share firms to enter the cross-country regressions and estimate the coefficient on the A-share indicator (row 1), we then allow incumbent firms and incumbent plus newly listed firms to enter the regressions (rows 2 and 3). In Panel C, we take the difference in the regression coefficients on the A-share indicator in Panel B to measure the marginal contribution of each of the three groups of firms to the overall performance gap. The contribution of the ST firms (newly listed firms) is measured as the difference in coefficients in row 1 and row 3 (rows 2 and row 3) of Panel B. We divide the performance gap of each group of firms by the total A-share performance gap to obtain the relative contribution (in percentages) of each group. When the dependent variable is stock returns, we employ WLS regressions in which we use lagged one-year market capitalization as the weight; when the dependent variable is an operating performance measure, we use OLS regressions. *t*-Statistics calculated using standard errors clustered by industry and year are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variables definitions.

Panel A. Performance of Large A-Share Firms and SOEs						
	Stock Returns (1)	Investment (2)	OCF (3)	Net Cash Flows (4)		
Large	-6.353*** (-3.952)	0.459*** (6.030)	-0.276 (-1.342)	-0.766*** (-3.752)		
SOE	-2.793* (-1.842)	0.074 (1.053)	0.138 (0.935)	0.132 (0.915)		
Year/Industry/Province FE	Yes	Yes	Yes	Yes		
Firm Controls	Yes	Yes	Yes	Yes		
R ² (%)	52.48	16.01	10.23	4.55		
Observations	36,044	35,879	34,459	34,459		
Differences in Coefs. of "Large" and "SOE"	-3.560 (-1.601)	0.385*** (3.717)	-0.414 (-0.405)	-0.898*** (-3.592)		
Panel B. Decompose Underperformance of A-Share Firms: Regression Coefficients of the A-Share Indicator						
	All Firms (1)	SOEs (2)	Non-SOEs (3)			
All A-share Firms (Incumbent + Newly Listed + ST)	-14.948** (-2.389)	-12.778*** (-3.645)	-12.746*** (-3.634)			
Incumbent Only	-7.274** (-2.192)	-7.596** (-2.038)	-5.373* (-1.655)			
Incumbent + Newly Listed	-11.893*** (-2.965)	-9.91*** (-2.834)	-10.167*** (-2.929)			
Panel C. Decompose Underperformance of A-Share Listed Firms (in Percentages)						
	All Firms (1)	All Firms (in %) (2)	SOEs (3)	SOEs (in %) (4)	Non-SOEs (5)	Non-SOEs (in %) (6)
Firms in ST Status	3.055	20.44%	2.868	22.44%	2.579	20.23%
Incumbent Firms	7.274**	48.66%	7.596**	59.45%	5.373*	42.15%
Newly Listed Firms	4.619	30.90%	2.314	18.11%	4.794	37.61%

Table XIII
Performance of Cross-Listed A-Share Firms

This table compares stock returns and operating performance of firms cross-listed in the A-share and Hong Kong markets, and non-cross-listed A-share firms. Panel A compares RPT activities of cross-listed and non-cross listed A-share firms. The dependent variables are *RPT Amt Out* and *Stock Returns* (in percentage points). Columns (1) and (3) show regression results of the full sample of A-share firms, while columns (2) and (4) show results of a matched sample of cross-listed and non-cross-listed A-share firms. Each cross-listed firm is matched with one non-cross-listed A-share firm in the same industry and with the closest book assets in the IPO year; 99 firms cross-listed in HKEX are matched following this procedure. Panel B presents results on operating performance. Explanatory variables are lagged by one year. In Panel A, columns (3) and (4), we employ WLS regressions, where the dependent variable is the stock return, and we use lagged one-year market capitalization as the weight. We employ OLS regressions in columns (1) and (2) of Panel A, and all columns in Panel B. *t*-Statistics calculated using standard errors clustered by industry and year are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels. Appendix A provides detailed variables definitions.

Panel A. Cross-Listing, RPT, and Stock Returns				
	RPT Amt Out		Stock Returns	
	Full Sample (1)	Matched Sample (2)	Full Sample (3)	Matched Sample (4)
Cross-Listed	-0.026** (-2.445)	0.001 (0.266)	1.588 (0.972)	-0.954 (-0.609)
Year/Industry/Location FE	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes
R ² (%)	4.67	17.66	46.72	62.23
Observations	33,028	2,306	38,505	2,423

Panel B. Cross-Listing and Operating Performance for Matched Sample						
	ROA (1)	ROE (2)	Net Income Growth (3)	Operating Cash Flows (4)	Investment (5)	Net Cash Flows (6)
Cross-Listed	-0.006 (-1.513)	0.033 (0.372)	-0.041 (-0.592)	-0.007 (-1.542)	0.007 (1.552)	-0.008 (-1.637)
Year/Industry/Location FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm Controls	Yes	Yes	Yes	Yes	Yes	Yes
R ² (%)	56.10	1.92	14.01	32.20	58.41	9.48
Observations	2,438	2,463	2,463	2,162	2,451	2,162

6 Short summary

- **Method.** Build a simple model of reform expectations, investor heterogeneity, and financial repression; then test its predictions using cross-country, within-China, IPO, governance, and sentiment evidence.
- **Identification.** The paper does not use one clean instrument; instead it relies on multiple comparison groups and outcome variables to separate institutional, behavioral, and discount-rate explanations.
- **Application.** China's A-share market versus externally listed Chinese firms, global listed firms, and matched unlisted Chinese firms over 1991–2018, with most tests focused on 2000–2018.
- **Main result.** A-share underperformance is best explained by a combination of **institutional deficiencies** and **behavioral biases**, with **financial repression** playing only a partial role. Weak governance helps explain both low returns and weak cash-flow performance; sentiment mainly explains low returns.

7 Conclusion

7.1 Core conclusion

- The Chinese A-share market performs poorly not because China as an economy performs poorly, but because the domestic listed market has a distorted mix of listed firms, weak exit discipline,

weak governance, and sentiment-driven pricing.

- Externally listed Chinese firms and matched unlisted Chinese firms both outperform A-share firms along important dimensions.
- The underperformance is concentrated among **large A-share firms**.

7.2 Economic intuition

- If listing is hard for the wrong reasons, the market may select firms that are **good at passing listing hurdles** rather than firms with the best long-run cash-flow prospects.
- If delisting is weak, poor firms remain in the market and listed shells acquire option value unrelated to productive performance.
- If governance is weak, firms can invest heavily without creating much shareholder value.
- If optimistic investors dominate pricing, returns can stay low for a long time even when future reforms never arrive.

7.3 Takeaway

This paper is best read as a broad diagnosis of why a very large stock market can fail to deliver either good long-run returns or strong operating performance. The central lesson is that **market size is not market quality**. To improve the A-share market, China needs not only deeper capital markets, but also **better entry rules, better exit rules, better governance, and a more rational investor base**.